

Muhao Chen

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RESEARCH INTERESTS

Natural Language Processing; Robustness in Machine Learning; AI Safety; Multi-modal and Agentic AI.

EDUCATION AND ACADEMIC EXPERIENCE

University of California, Davis, CA

- *Associate Professor (with tenure; step 3)*, Department of Computer Science 2026.7 - +
- *Applied Math (By Courtesy)*
- *Assistant Professor*, Department of Computer Science 2023.11-2026.6

University of Southern California, Los Angeles, CA

- *Adjunct Assistant Research Professor*, Department of Computer Science 2023.11-2026.7
- *Assistant Research Professor*, Department of Computer Science 2020.9-2023.10

University of Pennsylvania, Philadelphia, PA

2019.7-2020.8

Postdoctoral Fellow, Hosted by Dan Roth, Eduardo D. Glandt Distinguished Professor of Computer and Information Science

University of California, Los Angeles, CA

Ph.D. in Computer Science

2014.9-2019.6

Dissertation: Multi-relational Representation Learning and Knowledge Acquisition

Advisors:

- Carlo Zaniolo, *Distinguished Professor of Computer Science, N.E. Friedmann Chair in Knowledge Science* (Committee Chair)
- Kai-Wei Chang, *Associate Professor of Computer Science*
- Wei Wang, *Leonard Kleinrock Chair Professor of Computer Science*

Fudan University, Shanghai, China

B.S. in Computer Science. Advisor: X. Sean Wang, *Dorothean Chair Professor of Computer Science*

2010.9-2014.6

PUBLICATION

*Indicating equal contributions.

Refereed Publication in Conference Proceedings

- P1 Wenjie Jacky Mo, Qin Liu, Xiaofei Wen, Wenxuan Zhou, Zhe Zhao, **Muhao Chen**. DebugLM: Learning Traceable Training Data Provenance for LLMs. In *the 41st Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
- P2 Siddharth Sai, Xiaofei Wen, **Muhao Chen**. Robust and Efficient Guardrails with Latent Reasoning. In *the 41st Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
- P3 Hadi Askari, Japinder Singh, Tenghao Huang, Dongwon Jung, Anshuman Chhabra, **Muhao Chen**. Avoidance Learning of Failure Modes Produces Robust Tool Planners. In *the 41st Conference on Empirical Methods in Natural Language Processing (EMNLP) - Findings*, 2026.
- P4 Xiaofei Wen, Wenjie Jacky Mo, Yanan Xie, Peng Qi, **Muhao Chen**. Learning Efficient Guardrails for Compliance. In *the International Conference on Machine Learning (ICML)*, 2026.
- P5 Shahriar Kabir Nahin, Hadi Askari, **Muhao Chen**, Anshuman Chhabra. Less Diverse, Less Safe: The Indirect But Pervasive Risk of Test-Time Scaling in Large Language Models. In *the International Conference on Machine Learning (ICML)*, 2026.
- P6 Yichen Wang, Chenghao Yang, Tenghao Huang, **Muhao Chen**, Jonathan May, Mina Lee. Optimizing Diversity and Quality through Base-Aligned Model Collaboration. In *the International Conference on Machine Learning (ICML)*, 2026.
- P7 Dongwon Jung, Peng Shi, **Muhao Chen**, Yi Zhang. FutureWeaver: Planning Test-Time Compute for Multi-Agent Systems with Modularized Collaboration. *Third Conference on Language Modeling (CoLM)*, 2026.
- P8 Xiaogeng Liu, Xinyan Wang, Yechao Zhang, Sanjay Kariyappa, Chong Xiang, **Muhao Chen**, G Edward Suh, Chaowei Xiao. ReasoningBomb: A Stealthy Denial-of-Service Attack by Inducing Pathologically Long Reasoning in Large Reasoning Models. In *the ACM Conference on Computer and Communications Security (ACM CCS)*, 2026.

- P9 Wenjie Jacky Mo, Qin Liu, Xiaofei Wen, Dongwon Jung, Hadi Askari, Wenxuan Zhou, Zhe Zhao, **Muhao Chen**. RedCoder: Automated Multi-Turn Red Teaming for Code LLMs. In *the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.
- P10 Cheng Wang, Qin Liu, Wenxuan Zhou, **Muhao Chen**. Taming Extreme Tokens: Covariance-Aware GRPO with Gaussian-Kernel Advantage Reweighting. In *the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.
- P11 Tenghao Huang, Kung-Hsiang Huang, Prafulla Kumar Choubey, Yilun Zhou, **Muhao Chen**, Jonathan May, Chien-Sheng Wu. GTA: Generating Long-horizon Tasks for Web Agents at Scale. In *the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.
- P12 Qin Liu, Wenxuan Zhou, Nan Xu, James Y. Huang, Fei Wang, Sheng Zhang, Hoifung Poon, **Muhao Chen**. MetaScale: Test-Time Scaling with Evolving Meta-Thoughts. In *the 64th Annual Meeting of the Association for Computational Linguistics (ACL)* - Findings, 2026.
- P13 Cheng Wang, Zeming Wei, Qin Liu, **Muhao Chen**. False Sense of Security: Why Probing-based Malicious Input Detection Fails to Generalize. In *the 64th Annual Meeting of the Association for Computational Linguistics (ACL)* - Findings, 2026.
- P14 Qin Liu, Wenxuan Zhou, Nan Xu, James Y. Huang, Fei Wang, Sheng Zhang, Hoifung Poon, **Muhao Chen**.
- P15 Bangzheng Li, Ximeng Sun, Jiang Liu, Ze Wang, Jialian Wu, Xiaodong Yu, Hao Chen, Emad Barsoum, **Muhao Chen**, Zicheng Liu. Latent Visual Reasoning. *International Conference on Learning Representations (ICLR)*, 2026.
- P16 Kyle R. Chickering, Bangzheng Li, **Muhao Chen**. QLIP: A Dynamic Quadtree Vision Prior Enhances MLLM Performance Without Retraining. *International Conference on Learning Representations (ICLR)*, 2026.
- P17 Jiyeon Pyo, Yuankun Jiao, Dongwon Jung, Zekun Li, Leeje Jang, Sofia Kirsanova, Jina Kim, Yijun Lin, Qin Liu, Junyi Xie, Hadi Askari, Nan Xu, **Muhao Chen**, Yao-Yi Chiang. FRIEDA: Benchmarking Multi-Step Cartographic Reasoning in Vision-Language Models. *International Conference on Learning Representations (ICLR)*, 2026.
- P18 Bonan Kou, Zijie Zhou, **Muhao Chen**, Tianyi Zhang. Automating API Documentation from Crowdsourced Knowledge. In *the 48th IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026.
- P19 Qin Liu, Fei Wang, Chaowei Xiao, **Muhao Chen**. SudoLM: Learning Access Control of Parametric Knowledge with Authorization Alignment. In *the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.
- P20 Tenghao Huang, Kinjal Basu, Ibrahim Abdelaziz, Pavan Kapanipathi, Jonathan May, **Muhao Chen**. R2D2: Remembering, Replaying and Dynamic Decision Making with a Reflective Agentic Memory. In *the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.
- P21 Weidi Luo, Shenghong Dai, Xiaogeng Liu, Suman Banerjee, Huan Sun, **Muhao Chen**, Chaowei Xiao. AGrail: A Lifelong Agent Guardrail with Effective and Adaptive Safety Detection. In *the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.
- P22 Xiaofei Wen, Wenxuan Zhou, Wenjie Jacky Mo, **Muhao Chen**. ThinkGuard: Deliberative Slow Thinking Leads to Cautious Guardrails. In *the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)* - Findings, 2025.
- P23 Hadi Askari, Shivanshu Gupta, Fei Wang, Anshuman Chhabra, **Muhao Chen**. LayerIF: Estimating Layer Quality for Large Language Models using Influence Functions. In *the 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- P24 Fei Wang, Xingyu Fu, James Y. Huang, Zekun Li, Qin Liu, Xiaogeng Liu, Mingyu Derek Ma, Nan Xu, Wenxuan Zhou, Kai Zhang, Tianyi Lorena Yan, Wenjie Jacky Mo, Hsiang-Hui Liu, Pan Lu, Chunyuan Li, Chaowei Xiao, Kai-Wei Chang, Dan Roth, Sheng Zhang, Hoifung Poon, **Muhao Chen**. MuirBench: A Comprehensive Benchmark for Robust Multi-image Understanding. *International Conference on Learning Representations (ICLR)*, 2025.
- P25 Terry Tong, Fei Wang, Zhe Zhao, **Muhao Chen**. BadJudge: Backdoor Vulnerabilities of LLM-As-A-Judge. *International Conference on Learning Representations (ICLR)*, 2025.
- P26 Yingzi Ma, Jiong Xiao Wang, Fei Wang, Siyuan Ma, Jiazhao Li, Jinsheng Pan, Xiujun Li, Furong Huang, Lichao Sun, Bo Li, Yejin Choi, **Muhao Chen**, Chaowei Xiao. Benchmarking Vision Language Model Unlearning via Fictitious Facial Identity Dataset. *International Conference on Learning Representations (ICLR)*, 2025.
- P27 Dongwon Jung, Wenxuan Zhou, **Muhao Chen**. Code Execution as Grounded Supervision for LLM Reasoning. In *the 40th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
- P28 Hongxiang Zhang, Hao Chen, **Muhao Chen**, Tianyi Zhang. Active Layer-Contrastive Decoding Reduces Hallucination in Large Language Model Generation. In *the 40th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

- P29 Dongwon Jung, Qin Liu, Tenghao Huang, Ben Zhou, **Muhao Chen**. Familiarity-Aware Evidence Compression for Retrieval-Augmented Generation. In *the 40th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2025.
- P30 Tenghao Huang, Sihao Chen, **Muhao Chen**, Jonathan May, Longqi Yang, Mengting Wan, Pei Zhou. Teaching Language Models To Gather Information Proactively. In *the 40th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2025.
- P31 Jacob Dineen, Aswin RRV, Qin Liu, Zhikun Xu, Xiao Ye, Ming Shen, Zhaonan Li, Shijie Lu, Chitta Baral, **Muhao Chen**, Ben Zhou. QA-LIGN: Aligning LLMs through Constitutionally Decomposed QA. In *the 40th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2025.
- P32 Nan Xu, Fei Wang, Sheng Zhang, Hoifung Poon, **Muhao Chen**. From Introspection to Best Practices: Principled Analysis of Demonstrations in Multimodal In-Context Learning. In *the 22nd Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2025.
- P33 Wenjie Jacky Mo, Jiashu Xu, Qin Liu, Jiong Xiao Wang, Jun Yan, Hadi Askari, Chaowei Xiao, **Muhao Chen**. Test-time Backdoor Mitigation for Black-Box Large Language Models with Defensive Demonstrations. In *the 22nd Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)* - Findings, 2025.
- P34 Yiwei Wang, **Muhao Chen**, Nanyun Peng, Kai-Wei Chang. Vulnerability of Large Language Models to Output Prefix Jailbreaks: Impact of Positions on Safety. In *the 22nd Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)* - Findings, 2025.
- P35 Hadi Askari, Anshuman Chhabra, **Muhao Chen**, Prasant Mohapatra. Assessing LLMs for Zero-shot Abstractive Summarization Through the Lens of Relevance Paraphrasing. In *the 22nd Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)* - Findings, 2025.
- P36 Zilan Wang, Junfeng Guo, Jiacheng Zhu, Yiming Li, Heng Huang, **Muhao Chen**, Zhengzhong Tu. SleeperMark: Towards Robust Watermark against Fine-Tuning Text-to-image Diffusion Models. In *the IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR)*, 2025.
- P37 Tenghao Huang, Donghee Lee, John Sweeney, Jiatong Shi, Emily Steliotes, Matthew Lange, Jonathan May, **Muhao Chen**. Food-Puzzle: Developing Large Language Model Agents as Flavor Scientists. In *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)* - Datasets and Benchmarks Track, 2025.
- P38 Jianpeng Chen, Wangzhi Zhan, Haohui Wang, Zian Jia, Jingru Gan, Junkai Zhang, Jingyuan Qi, Tingwei Chen, Lifu Huang, **Muhao Chen**, Ling Li, Wei Wang, Dawei Zhou. In *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)* - Datasets and Benchmarks Track, 2025.
- P39 Zekun Li, Malcolm Grossman, Eric Qasemi, Mihir Kulkarni, **Muhao Chen**, Yao-Yi Chiang. Benchmarking Geospatial Question Answering with MapQA. In *the 33rd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)* - Applications Track, 2025.
- P40 Wenxuan Zhou, Sheng Zhang, Yu Gu, **Muhao Chen**, Hoifung Poon. UniversalNER: Targeted Distillation from Large Language Models for Open Named Entity Recognition. *International Conference on Learning Representations (ICLR)*, 2024.
- P41 Xiaogeng Liu, Nan Xu, **Muhao Chen**, Chaowei Xiao. AutoDAN: Generating Stealthy Jailbreak Prompts on Aligned Large Language Models. *International Conference on Learning Representations (ICLR)*, 2024.
- P42 Qin Liu, Fei Wang, Chaowei Xiao, **Muhao Chen**. From Shortcuts to Triggers: Backdoor Defense with Denoised PoE. In the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2024.
- P43 Bangzheng Li, Ben Zhou, Fei Wang, Xingyu Fu, Dan Roth, **Muhao Chen**. Deceptive Semantic Shortcuts on Reasoning Chains: How Far Can Models Go without Hallucination? In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- P44 Jiashu Xu, Mingyu Derek Ma, Fei Wang, Chaowei Xiao, **Muhao Chen**. Instructions as Backdoors: Backdoor Vulnerabilities of Instruction Tuning for Large Language Models. In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- P45 Jiashu Xu, Fei Wang, Mingyu Derek Ma, Pang Wei Koh, Chaowei Xiao, **Muhao Chen**. Instructional Fingerprinting of Large Language Models. In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- P46 Tianyang Liu, Fei Wang, **Muhao Chen**. Rethinking Tabular Data Understanding with Large Language Models. In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- P47 Victoria Graf, Qin Liu, **Muhao Chen**. Two Heads are Better than One: Nested PoE for Robust Defense Against Multi-Backdoors. In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

- P48 Lingbo Mo, Boshi Wang, **Muhao Chen**, Huan Sun. How Trustworthy are Open-Source LLMs? An Assessment under Malicious Demonstrations Shows their Vulnerabilities. In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- P49 Tenghao Huang, Dongwon Jung, Vaibhav Kumar, Mohammad Kachuee, Xiang Li, Puyang Xu, **Muhao Chen**. Planning and Editing What You Retrieve for Enhanced Tool Learning. In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) - Findings*, 2024.
- P50 Nan Xu, Fei Wang, Ben Zhou, Bangzheng Li, Chaowei Xiao, **Muhao Chen**. Cognitive Overload: Jailbreaking Large Language Models with Overloaded Logical Thinking. In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) - Findings*, 2024.
- P51 Tianqing Fang, Zhaowei Wang, Wenxuan Zhou, Hongming Zhang, Yangqiu Song, **Muhao Chen**. Getting Sick After Seeing a Doctor? Diagnosing and Mitigating Knowledge Conflicts in Event Temporal Reasoning. In *the 21st Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) - Findings*, 2024.
- P52 Jiongxiao Wang, Junlin Wu, **Muhao Chen**, Yevgeniy Vorobeychik, Chaowei Xiao. On the Exploitability of Reinforcement Learning with Human Feedback for Large Language Models. In *the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- P53 Tianyi Yan, Fei Wang, James Y. Huang, Wenxuan Zhou, Fan Yin, Aram Galstyan, Wenpeng Yin, **Muhao Chen**. Contrastive Instruction Tuning. In *the 62nd Annual Meeting of the Association for Computational Linguistics (ACL) - Findings*, 2024.
- P54 Hanzi Xu, **Muhao Chen**, Lifu Huang, Slobodan Vucetic, Wenpeng Yin. X-Shot: A Unified System to Handle Frequent, Few-shot and Zero-shot Learning Simultaneously in Classification. In *the 62nd Annual Meeting of the Association for Computational Linguistics (ACL) - Findings*, 2024.
- P55 Fei Wang, Wenxuan Zhou, James Y. Huang, Nan Xu, Sheng Zhang, Hoifung Poon, **Muhao Chen**. mDPO: Conditional Preference Optimization for Multimodal Large Language Models. In *the 39th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
- P56 Xiaofei Wen, Bangzheng Li, Tenghao Huang, **Muhao Chen**. Red Teaming Language Models for Processing Contradictory Dialogues. In *the 39th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
- P57 Yufei Tian, Tenghao Huang, Miri Liu, Derek Jiang, Alexander Spangher, **Muhao Chen**, Jonathan May, Nanyun Peng. Are Large Language Models Capable of Generating Human-Level Narratives? In *the 39th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024. **Outstanding Paper Award**
- P58 Qin Liu, Fei Wang, Nan Xu, Tianyi Lorena Yan, Tao Meng, **Muhao Chen**. Monotonic Paraphrasing Improves Generalization of Language Model Prompting. In *the 39th Conference on Empirical Methods in Natural Language Processing (EMNLP) - Findings*, 2024.
- P59 Terry Tong, Qin Liu, Jiashu Xu, **Muhao Chen**. Securing Multi-turn Conversational Language Models from Distributed Backdoor Attacks. In *the 39th Conference on Empirical Methods in Natural Language Processing (EMNLP) - Findings*, 2024.
- P60 Jiongxiao Wang, Jiazhao Li, Yiquan Li, Xiangyu Qi, Junjie Hu, Yixuan Li, Patrick McDaniel, **Muhao Chen**, Bo Li, Chaowei Xiao. Mitigating Fine-tuning based Jailbreak Attack with Backdoor Enhanced Safety Alignment. In *the 38th Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024
- P61 Yu Wang, Xiaogeng Liu, Yu Li, **Muhao Chen**, Chaowei Xiao. AdaShield: Safeguarding Multimodal Large Language Models from Structure-based Attack via Adaptive Shield Prompting. *The 18th European Conference on Computer Vision (ECCV)*, 2024.
- P62 Tianqing Fang, Wenxuan Zhou, Fangyu Liu, Hongming Zhang, Yangqiu Song, **Muhao Chen**. On-the-fly Denoising for Data Augmentation in Natural Language Understanding. In *the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL) - Findings*, 2024.
- P63 Qin Liu, Wenjie Mo, Terry Tong, Jiashu Xu, Fei Wang, Chaowei Xiao, **Muhao Chen**. Mitigating Backdoor Threats to Large Language Models: Advancement and Challenges. In *Proceedings of the 60th Annual Allerton Conference (Allerton)*, 2024. **Invited Paper**
- P64 James Y. Huang, Wenlin Yao, Kaiqiang Song, Hongming Zhang, **Muhao Chen**, Dong Yu. Bridging Continuous and Discrete Spaces: Interpretable Sentence Representation Learning via Compositional Operations. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. **Outstanding Paper Award**
- P65 Haoyu Wang, Hongming Zhang, Yueguan Wang, Yuqian Deng, **Muhao Chen**, Dan Roth. Are All Steps Equally Important? Benchmarking Essentiality Detection of Events. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

- P66 Zekun Li, Wenxuan Zhou, Yao-Yi Chiang, **Muhao Chen**. GeoLM: Empowering Language Models for Geospatially Grounded Language Understanding. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- P67 Yiwei Wang, Yujun Cai, **Muhao Chen**, Yuxuan Liang, Bryan Hooi. Primacy Effect of ChatGPT. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- P68 Fei Wang, Wenjie Mo, Yiwei Wang, Wenxuan Zhou, **Muhao Chen**. A Causal View of Entity Bias in (Large) Language Models. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP) - Findings*, 2023.
- P69 Wenxuan Zhou, Sheng Zhang, Hoifung Poon, **Muhao Chen**. Context-faithful Prompting for Large Language Models. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP) - Findings*, 2023.
- P70 Nan Xu, Fei Wang, Mingtao Dong, **Muhao Chen**. Dense Retrieval as Indirect Supervision for Large-space Decision Making. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP) - Findings*, 2023.
- P71 Tenghao Huang, Ehsan Qasemi, Bangzheng Li, He Wang, Faeze Brahman, **Muhao Chen**, Snigdha Chaturvedi. Affective and Dynamic Beam Search for Story Generation. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP) - Findings*, 2023.
- P72 Shikhar Singh, Ehsan Qasemi, **Muhao Chen**. VIPHY: Probing “Visible” Physical Commonsense Knowledge. In *the 38th Conference on Empirical Methods in Natural Language Processing (EMNLP) - Findings*, 2023.
- P73 Yiwei Wang, Bryan Hooi, Fei Wang, Yujun Cai, Yuxuan Liang, Wenxuan Zhou, Jing Tang, Manjuan Duan, **Muhao Chen**. How Fragile is Relation Extraction under Entity Replacements? In *the 27th SIGNLL Conference on Computational Natural Language Learning (CoNLL)*, 2023.
- P74 Wenxuan Zhou, Sheng Zhang, Tristan Naumann, **Muhao Chen**, Hoifung Poon. Continual Contrastive Finetuning Improves Low-Resource Relation Extraction. In *the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- P75 Jiashu Xu, Mingyu Derek Ma, **Muhao Chen**. Can NLI Provide Proper Indirect Supervision for Low-resource Biomedical Relation Extraction? In *the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- P76 Tanay Dixit, Fei Wang, **Muhao Chen**. Improving Factuality of Abstractive Summarization without Sacrificing Summary Quality. In *the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- P77 Shudi Hou, Yu Xia, **Muhao Chen**, Sujian Li. Contrastive Bootstrapping for Label Refinement. In *the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- P78 Fei Wang*, James Y. Huang*, Tianyi Yan, Wenxuan Zhou, **Muhao Chen**. Robust Natural Language Understanding with Residual Attention Debiasing. In *the 61st Annual Meeting of the Association for Computational Linguistics (ACL) - Findings*, 2023.
- P79 Keming Lu, I-Hung Hsu, Wenxuan Zhou, Mingyu Derek Ma, **Muhao Chen**. Multi-hop Evidence Retrieval for Cross-document Relation Extraction. In *the 61st Annual Meeting of the Association for Computational Linguistics (ACL) - Findings*, 2023.
- P80 Xinze Li, Yixin Cao, **Muhao Chen**, Aixin Sun. Take a Break in the Middle: Investigating Subgoals towards Hierarchical Script Generation. In *the 61st Annual Meeting of the Association for Computational Linguistics (ACL) - Findings*, 2023.
- P81 Haoyu Wang, Hongming Zhang, Yuqian Deng, Jacob Gardner, Dan Roth, **Muhao Chen**. Extracting or Guessing? Improving Faithfulness of Event Temporal Relation Extraction. In *the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2023.
- P82 Xiaocong Yang, James Y. Huang, Wenxuan Zhou, **Muhao Chen**. Parameter-Efficient Tuning with Special Token Adaptation. In *the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2023.
- P83 Peifeng Wang, Aaron Chan, Filip Ilievski, **Muhao Chen**, Xiang Ren. PINTO: Faithful Language Reasoning Using Prompted-Generated Rationales. In *the 11th International Conference on Learning Representations (ICLR)*, 2023.
- P84 Bonan Kou, **Muhao Chen**, Tianyi Zhang. Automated Summarization of Stack Overflow Posts. In *the Proceedings of the 45th IEEE/ACM International Conference on Software Engineering (ICSE)*, 2023.
- P85 Zhongkai Zhao, Bonan Kou, Mohamed Yilmaz Ibrahim, **Muhao Chen**, Tianyi Zhang. Knowledge-based Version Incompatibility Detection for Deep Learning. In *the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2023.
- P86 Tai Nguyen, Yifeng Di, Joohan Lee, **Muhao Chen**, Tianyi Zhang. Software Entity Recognition with Noise-Robust Learning. In *the Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2023.
- P87 Minh Pham, Craig Knoblock, **Muhao Chen**. Detecting Semantic Errors in Tables using Textual Evidence. In *the Proceedings of IEEE International Conference on Big Data (BigData)*, 2023.

- P88 Wenxuan Zhou, Fangyu Liu, Huan Zhang, **Muhao Chen**. Sharpness-Aware Minimization with Dynamic Reweighting. In *the 37th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- P89 Nan Xu, Fei Wang, Bangzheng Li, Mingtao Dong, **Muhao Chen**. Does Your Model Classify Entities Reasonably? Diagnosing and Mitigating Spurious Correlations in Entity Typing. In *the 37th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- P90 Fei Wang, Kaiqiang Song, Hongming Zhang, Lifeng Jin, Sangwoo Cho, Wenlin Yao, Xiaoyang Wang, **Muhao Chen**, Dong Yu. Saliency Allocation as Guidance for Abstractive Summarization. In *the 37th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- P91 Keming Lu, I-Hung Hsu, Wenxuan Zhou, Mingyu Derek Ma, **Muhao Chen**. Summarization as Indirect Supervision for Relation Extraction. In *the 37th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2022.
- P92 Zekun Li, Jina Kim, Yao-Yi Chiang, **Muhao Chen**. SpaBERT: Pretrained Language Models on Geographic Data for Geo-Entity Representation. In *the 37th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2022.
- P93 Ehsan Qasemi, Filip Ilievski, **Muhao Chen**, Pedro Szekely. PaCo: Preconditions Attributed to Commonsense Knowledge. In *the 37th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2022.
- P94 Yiwei Wang, **Muhao Chen**, Wenxuan Zhou, Yujun Cai, Yuxuan Liang, Dayiheng Liu, Baosong Yang, Juncheng Liu, Bryan Hooi. Should We Rely on Entity Mentions for Relation Extraction? Debiasing Relation Extraction with Counterfactual Analysis. In *the 20th Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2022.
- P95 Fei Wang, Zhewei Xu, Pedro Szekely, **Muhao Chen**. Robust (Controlled) Table-to-Text Generation with Structure-Aware Equivariance Learning. In *the 20th Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2022.
- P96 James Y. Huang, Bangzheng Li, Jiashu Xu, **Muhao Chen**. Unified Semantic Typing with Meaningful Label Inference. In *the 20th Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2022.
- P97 Wenxuan Zhou, Qiang Ning, Heba Elfardy, Kevin Small, **Muhao Chen**. Answer Consolidation: Formulation and Benchmarking. In *the 20th Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2022.
- P98 Juncheng Liu, Zequn Sun, Bryan Hooi, Yiwei Wang, Dayiheng Liu, Baosong Yang, Xiaokui Xiao, **Muhao Chen**. Dangling-Aware Entity Alignment with Mixed High-Order Proximities. In *the 20th Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)* - Findings, 2022.
- P99 Yiwei Wang, **Muhao Chen**, Wenxuan Zhou, Yujun Cai, Yuxuan Liang, Bryan Hooi. GraphCache: Message Passing as Caching for Sentence-Level Relation Extraction. In *the 20th Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)* - Findings, 2022.
- P100 Wenxuan Zhou*, Fangyu Liu*, Ivan Vulić, Nigel Collier, **Muhao Chen**. Prix-LM: Pretraining for Multilingual Knowledge Base Construction. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.
- P101 Peifeng Wang, Jonathan Zamora, Junfeng Liu, Fangyu Liu, **Muhao Chen**, Xiang Ren. Contextualized Scene Imagination for Generative Commonsense Reasoning. In *the 10th International Conference on Learning Representations (ICLR)*, 2022.
- P102 Wenxuan Zhou, **Muhao Chen**. An Improved Baseline for Sentence-level Relation Extraction. In *the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)*, 2022.
- P103 Ehsan Qasemi, Piyush Khanna, Qiang Ning, **Muhao Chen**. PInKS: Preconditioned Commonsense Inference with Minimal Supervision. In *the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)*, 2022.
- P104 Jihoon Sohn, Mingyu Derek Ma, **Muhao Chen**. Bending the Future: Autoregressive Modeling of Temporal Knowledge Graphs in Curvature-Variable Hyperbolic Spaces. In *the 4th Conference on Automated Knowledge Base Construction (AKBC)*, 2022.
- P105 Bonan Kou, Yifeng Di, **Muhao Chen**, Tianyi Zhang. SOSum: A Dataset of Stack Overflow Post Summaries. In *Proceedings of the 19th International Conference on Mining Software Repositories (MSR)*, 2022. (Data/Tool Showcase Track)
- P106 Wenxuan Zhou, Fangyu Liu, **Muhao Chen**. Contrastive Out-of-Distribution Detection for Pretrained Transformers. In *Proceedings of the 36th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
- P107 Wenxuan Zhou, **Muhao Chen**. Learning from Noisy Labels for Entity-Centric Information Extraction. In *Proceedings of the 36th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
- P108 Haoyu Wang, Hongming Zhang, **Muhao Chen**, Dan Roth. Learning Constraints and Descriptive Segmentation for Subevent Detection. In *Proceedings of the 36th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
- P109 Xiyang Zhang, **Muhao Chen**, Jonathan May. Saliency-Aware Event Chain Modeling for Narrative Understanding. In *Proceedings of the 36th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

- P110 Fei Wang, Kexuan Sun, Jay Pujara, Pedro Szekely, **Muhao Chen**. Table-based Fact Verification With Saliency-aware Learning. In *the 36th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2021.
- P111 Mingyu Derek Ma, **Muhao Chen**, Te-lin Wu, Nanyun Peng. HyperExpan: Taxonomy Expansion with Hyperbolic Representation Learning. In *the 36th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2021.
- P112 Fangyu Liu, **Muhao Chen**, Dan Roth, Nigel Collier. Visual Pivoting for (Unsupervised) Entity Alignment. In *the 35th AAAI Conference on Artificial Intelligence (AAAI)*, 2021.
- P113 Cunchao Zhu, **Muhao Chen**, Changjun Fan, Guangquan Cheng, Yan Zhang. Learning from History: Modeling Temporal Knowledge Graphs with Sequential Copy-Generator Networks. In *the 35th AAAI Conference on Artificial Intelligence (AAAI)*, 2021.
- P114 Zequn Sun, **Muhao Chen**, Wei Hu. Knowing the No-match: Entity Alignment with Dangling Cases. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.
- P115 Peifeng Wang, Filip Ilievski, Muhao Chen, Xiang Ren. Do Language Models Perform Generalizable Commonsense Inference? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)* - Findings, 2021.
- P116 **Muhao Chen**, Weijia Shi, Ben Zhou, Dan Roth. Cross-lingual Entity Alignment with Incidental Supervision. In *the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2021.
- P117 Xuelu Chen*, Michael Boratko*, **Muhao Chen**, Shib Sankar Dasgupta, Xiang Li, Andrew McCallum. Probabilistic Box Embeddings for Uncertain Knowledge Graph Reasoning. In *the 19th Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2021.
- P118 Fei Wang, Kexuan Sun, **Muhao Chen**, Jay Pujara, Pedro Szekely. Retrieving Complex Tables with Multi-Granular Graph Representation Learning. In *Proceedings of the 44th ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2021.
- P119 Minh Pham, Craig Knoblock, **Muhao Chen**, Binh Vu, Jay Pujara. SPADE: A Semi-supervised Probabilistic Approach for Detecting Errors in Tables. In *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.
- P120 Kexuan Sun, Fei Wang, **Muhao Chen**, Jay Pujara. Tabular Functional Block Detection with Embedding-based Agglomerative Cell Clustering. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021.
- P121 **Muhao Chen**, Hongming Zhang, Haoyu Wang, Dan Roth. “What Are You Trying to Do?” Semantic Typing of Event Processes. In *Proceedings of the 24th SIGNLL Conference on Computational Natural Language Learning (CoNLL)*, 2020. **Best Paper Nomination**
- P122 Zequn Sun, **Muhao Chen**, Wei Hu, Chengming Wang. Knowledge Association with Hyperbolic Representation Learning of Knowledge Graphs. In *Proceedings of the 25th Conference on Empirical Methods in Natural Language Processing (EMNLP)* 2020.
- P123 Haoyu Wang, **Muhao Chen**, Hongming Zhang, Dan Roth. Joint Constrained Learning for Event-Event Relation Extraction. In *Proceedings of the 25th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020.
- P124 Hongming Zhang, **Muhao Chen**, Haoyu Wang, Y. Song, Dan Roth. Analogous Process Structure Induction for Sub-event Sequence Prediction. In *Proceedings of the 25th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020.
- P125 Xuelu Chen, **Muhao Chen**, Changjun Fan, Ankith Uppunda, Yizhou Sun, Carlo Zaniolo. Multilingual Knowledge Graph Completion via Ensemble Knowledge Transfer. In *Proceedings of the 25th Conference on Empirical Methods in Natural Language Processing (EMNLP)* - Findings, 2020.
- P126 Zequn Sun, Chengming Wang, Wei Hu, **Muhao Chen**, Jian Dai, Wei Zhang, Yuzhong Qu. Knowledge Graph Alignment Network with Gated Multi-hop Neighborhood Aggregation. In *the 34th AAAI Conference on Artificial Intelligence (AAAI)*, 2020.
- P127 Changping Meng, **Muhao Chen**, Jie Mao, Jennifer Neville. ReadNet: A Hierarchical Transformer Framework for Readability Analysis. In *the 42nd European Conference on Information Retrieval (ECIR)*, 2020.
- P128 Junheng Hao, Chelsea J. T. Ju, **Muhao Chen**, Yizhou Sun, Carlo Zaniolo, Wei Wang. Bio-JOIE: Joint Representation Learning of Biological Knowledge Bases. In *the 11th ACM SIGBio Conference on Bioinform., Comput. Bio. and Health Inform. (ACM-BCB)*, 2020. **SIGBio Best Student Paper Award**
- P129 Tianran Zhang, **Muhao Chen**, Alex Bui. Diagnostic Prediction with Sequence-of-sets Representation Learning for Clinical Events. In *Proceedings of the 18th International Conference on Artificial Intelligence in Medicine (AIME)*, 2020
- P130 **Muhao Chen**, Yingtao Tian, Haochen Chen, Kai-Wei Chang, Steve Skiena, Carlo Zaniolo. Learning to Represent Bilingual Dictionaries. In *Proceedings of the 23rd SIGNLL Conference on Computational Natural Language Learning (CoNLL)*, 2019
- P131 Junheng Hao, **Muhao Chen**, Wenchao Yu, Yizhou Sun, Wei Wang. Universal Representation Learning of Knowledge Bases by Jointly Embedding Ontological Concepts and Instances. In *Proceedings of the 25th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2019.

- P132 Xuelu Chen, **Muhao Chen**, Weijia Shi, Yizhou Sun, Carlo Zaniolo. Uncertain Knowledge Graphs Embeddings. In *the 33rd International Conference on Artificial Intelligence (AAAI)*, 2019.
- P133 **Muhao Chen**^{*}, Weijia Shi^{*}, Pei Zhou, Kai-Wei Chang. Retrofitting Contextualized Word Embeddings with Paraphrases. In *Proceedings of the 24th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- P134 Pei Zhou, Weijia Shi, Jieyu Zhao, Kuan-Hao Huang, **Muhao Chen**, Ryan Cotterell, Kai-Wei Chang. Examining Gender Bias in Languages with Grammatical Gender. In *Proceedings of the 24th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- P135 **Muhao Chen**, Chris Quirk. Embedding Edge-attributed Relational Hierarchies. In *Proceedings of the 42nd ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2019.
- P136 Qingheng Zhang, Zequn Sun, Wei Hu, **Muhao Chen**, Lingbing Guo, Yuzhong Qu. Multi-view Knowledge Graph Embedding for Entity Alignment. In *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*, 2019. Also in ISWC 2020 Invited Sister Conference Track.
- P137 Zequn Sun, Jiacheng Huang, Wei Hu, **Muhao Chen**, Yuzhong Qu. TransEdge: Translating Relation-contextualized Embeddings for Knowledge Graphs. In *the 18th International Semantic Web Conference (ISWC)*, 2019.
- P138 Haochen Chen, Syed Fahad Sultan, Yingtao Tian, **Muhao Chen**, Steven Skiena. Fast and Accurate Network Embeddings via Very Sparse Random Projection. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*, 2019.
- P139 Changjun Fan, Yuhui Ding, Li Zeng, **Muhao Chen**, Yizhou Sun and Zhong Liu. Learning to Identify High Betweenness Centrality Nodes from Scratch: A Novel Graph Neural Network Approach. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*, 2019.
- P140 **Muhao Chen**, Changping Meng, Gang Huang, Carlo Zaniolo. Learning to Differentiate Between Main-articles and Sub-articles in Wikipedia. In *IEEE International Conference on Big Data (BigData)*, 2019.
- P141 Yingtao Tian, Haochen Chen, Bryan Perozzi, **Muhao Chen**, Xiaofei Sun, Steven Skiena. Social Relation Inference via Label Propagation. In *the 41st European Conference on Information Retrieval (ECIR)*, 2019.
- P142 Qi Zhao, **Muhao Chen**, Pengyuan Du, Tuan Le, Mario Gerla. Towards Efficient Cellular Traffic Offloading via Dynamic MPTCP Path Configuration with SDN. In *IEEE International Conference on Computing, Networking and Communications (ICNC)*, 2019.
- P143 **Muhao Chen**, Gang Huang, Changping Meng, Carlo Zaniolo. Neural Article Pair Modeling for Wikipedia Sub-article Matching. In *the 29th European Conference on Machine Learning (ECML)*, 2018 (**Plenary Presentation**, ~1.7% acceptance rate)
- P144 **Muhao Chen**, Yingtao Tian, Kai-Wei Chang, Steven Skiena, Carlo Zaniolo. Co-training Embeddings of Knowledge Graphs and Entity Descriptions for Cross-lingual Entity Alignment. In *the 27th International Joint Conference on Artificial Intelligence (IJCAI)*, 2018.
- P145 **Muhao Chen**, Yingtao Tian, Xuelu Chen, Zijun Xue, Carlo Zaniolo. On2Vec: Embedding-based Relation Prediction for Ontology Population. In *Proceedings of the 17th SIAM International Conference on Data Mining (SDM)*. SIAM, 2018
- P146 Haochen Chen, Xiaofei Sun, Yingtao Tian, Bryan Perozzi, **Muhao Chen** and Steven Skiena. Enhanced Network Embeddings via Exploiting Edge Labels. In *the 27th ACM Conference on Information and Knowledge Management (CIKM)*. ACM 2018.
- P147 Pengyuan Du, Seunghyun Yoo, Qi Zhao, **Muhao Chen**, Mario Gerla. Towards Opportunistic Resource Sharing in Mobile Social Networks - an Evolutionary Game Theoretic Approach. In *Proceedings of the 19th ACM International Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc)*, ACM 2018.
- P148 **Muhao Chen**, Qi Zhao, Pengyuan Du, Carlo Zaniolo, Mario Gerla. Demand-driven Cache Allocation Based on Context-aware Collaborative Filtering. In *Proceedings of the 19th ACM International Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc)*, ACM 2018.
- P149 **Muhao Chen**, Yingtao Tian, Mohan Yang, Carlo Zaniolo. Multilingual Knowledge Graph Embeddings for Cross-lingual Knowledge Alignment. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*. 2017.
- P150 **Muhao Chen**, Shi Gao, X. Sean Wang. Converting Spatiotemporal Data Among Heterogeneous Granularity Systems. In *Proceedings of the 25th IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. IEEE, 2016.

Refereed Journal Publication and Book Chapters

- J1 James Y. Huang, Wenxuan Zhou, Fei Wang, Fred Morstatter, Sheng Zhang, Hoifung Poon, Muhao Chen. Offset Unlearning for Large Language Models. *Transactions on Machine Learning Research (TMLR)*. 2025
- J2 Juan Zambrano Chaves, Mr Shih-Cheng Huang, Yanbo Xu, Mr Hanwen Xu, Dr Naoto Usuyama, Dr Sheng Zhang, Fei Wang, Yujia Xie, Mahmoud Khademi, Ziyi Yang, Hany Awadalla, Julia Gong, Houdong Hu, Jianwei Yang, Chunyuan Li, Jianfeng Gao, Yu Gu,

- Cliff Wong, Mu Wei, Tristan Naumann, Muhao Chen, Matthew Lungren, Akshay Chaudhari, Serena Yeung-Levy, Curtis Langlotz, Sheng Wang, Hoifung Poon. A Clinically Accessible Small Multimodal Radiology Model and Evaluation Metric for Chest X-Ray Findings. **Nature Communications**. 16 (1). Nature Portfolio. 2025.
- J3 Yao-Yi Chiang, **Muhao Chen**, Weiwei Duan, Jina Kim, Craig Knoblock, Stefan Leyk, Zekun Li, Yijun Lin, Min Namgung, Basel Shbita, Johannes H. Uhl. GeoAI for the Digitization of Historical Maps. *Handbook of GeoAI*. Taylor & Francis, 2023 (Book Chapter)
- J4 Bangzheng Li, Wenpeng Yin, **Muhao Chen**. Ultra-fine Entity Typing with Indirect Supervision from Natural Language Inference. *Transactions of the Association for Computational Linguistics (TACL)*. MIT Press, 2022.
- J5 Tianran Zhang, **Muhao Chen**, Alex Bui. AdaDiag: Adversarial Domain Adaptation of Diagnostic Prediction with Clinical Event Sequences. *Journal of Biomedical Informatics (JBI)*, vol. 134. Elsevier, 2022.
- J6 Mohammad Rostami, Hangfeng He, **Muhao Chen**, Dan Roth. Transfer Learning via Representation Learning. *Federated and Transfer Learning*. Springer, 2022 (Book Chapter)
- J7 Jyun-Yu Jiang, Chelsea J.-T. Ju, Junheng Hao, **Muhao Chen**, Wei Wang. Circular RNA Prediction based on Junction Encoders and Deep Interaction among Splice Sites. **Bioinformatics**, vol. 37. Oxford University Press, 2021. Full Paper of **ISMB/ECCB**, 2021.
- J8 Guangyu Zhou*, **Muhao Chen***, Chelsea J. T. Ju*, Zheng Wang, Jyun-Yu Jiang, Wei Wang. Mutation effect estimation on protein-protein interactions using deep contextualized representation learning. **NAR Genom. Bioinform**, vol. 2 (2). Oxford University Press. 2020.
- J9 Zequn Sun, Qingheng Zhang, Wei Hu, Chengming Wang, **Muhao Chen**, Chengkai Li, Yuzhong Qu. A Benchmarking Study of Embedding-based Entity Alignment for Knowledge Graphs. *Proceedings of the VLDB Endowment (PVLDB)*, vol. 13. ACM. 2020
- J10 **Muhao Chen***, Chelsea J. T. Ju*, Guangyu Zhou, Tianran Zhang, Kai-Wei Chang, Carlo Zaniolo, Wei Wang. Multifaceted Protein-Protein Interaction Prediction Based on Siamese Residual RCNN. **Bioinformatics**, vol. 35 (14) Oxford University Press. Full Paper of **ISMB/ECCB**, 2019.
- J11 Carlo Zaniolo, Shi Gao, Maurizio Atzori, **Muhao Chen**, Jiaqi Gu. User-Friendly Temporal Queries on Historical Knowledge Bases. **Information and Computation**, Vol. 259 (3). Elsevier, 2018.

Tutorials

- T1 **Muhao Chen**, Chaowei Xiao, Huan Sun, Lei Li, Leon Derczynski, Anima Anandkumar, Fei Wang. Combating Security and Privacy Issues in the Era of Large Language Models. In **NAACL**, 2024.
- T2 Wenpeng Yin, **Muhao Chen**, Rui Zhang, Ben Zhou, Fei Wang, Dan Roth. Enhancing LLM Capabilities Beyond Scaling Up. In **EMNLP**, 2024.
- T3 Wenpeng Yin, **Muhao Chen**, Ben Zhou, Qiang Ning, Kai-Wei Chang, Dan Roth. Indirectly Supervised Natural Language Processing. In **ACL**, 2023.
- T4 **Muhao Chen**, Lifu Huang, Manling Li, Ben Zhou, Heng Ji, Dan Roth. New Frontiers of Information Extraction. In **NAACL**, 2022.
- T5 **Muhao Chen**, Hongming Zhang, Qiang Ning, Manling Li, Heng Ji, Kathleen McKeown, Dan Roth. Event-centric Natural Language Processing. In **ACL**, 2021.
- T6 **Muhao Chen**, Hongming Zhang, Qiang Ning, Manling Li, Heng Ji, Dan Roth. Event-centric Natural Language Understanding. In **AAAI**, 2021.
- T7 Jay Pujara, Pedro Szekely, Huan Sun, **Muhao Chen**. From Tables to Knowledge: Recent Advances in Table Understanding. In **KDD**, 2021.
- T8 **Muhao Chen**, Kai-Wei Chang, Dan Roth. Recent Advances in Transferable Representation Learning. In **AAAI**, 2020.

Refereed Workshop and System Demonstration Papers

- W1 Jingyuan Qi, Zian Jia, Minqian Liu, Wangzhi Zhan, Junkai Zhang, Xiaofei Wen, Jingru Gan, Jianpeng Chen, Qin Liu, Mingyu Derek Ma, Bangzheng Li, Haohui Wang, Adithya Kulkarni, Muhao Chen, Dawei Zhou, Ling Li, Wei Wang, Lifu Huang. MetaScientist: A Human-AI Synergistic Framework for Automated Mechanical Metamaterial Design. In *the 22nd Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2025. (demo)
- W2 Weijia Shi, **Muhao Chen**, Yingtao Tian, Kai-Wei Chang. Learning Bilingual Word Embeddings Using Lexical Definitions. In *Proceedings of ACL Workshop on Representation Learning for NLP (RepL4NLP)*, 2019.
- W3 Zhubo Deng, Pei Zhou, Weijia Shi, **Muhao Chen**, Kai-Wei Chang. Computational Analysis of French-origin Reborrowing Process for English Loanwords. In *ICDM Workshop on Multilingual Cognitive Services (ICDMW)*, 2019

- W4 Changjun Fan, Yizhou Sun, Li Zeng, Yang-Yu Liu, **Muhao Chen**, Zhong Liu. Dismantle Large Networks through Deep Reinforcement Learning. In *ICLR Workshops*, 2019.
- W5 Pei Zhou, **Muhao Chen**, Kai-Wei Chang, Carlo Zaniolo. Quantification and Analysis of Scientific Language Variation by Research Fields. In *Proceedings of the ICDM Workshops (ICDMW)*, 2018.
- W6 **Muhao Chen**, Tao Zhou, Pei Zhou, Carlo Zaniolo. Multi-graph Affinity Embeddings for Multilingual Knowledge Graphs. **Contributed talk** in the *6th Workshop on Automated Knowledge Base Construction at NIPS (AKBC)*. 2017.
- W7 **Muhao Chen**, Carlo Zaniolo. Learning Multi-faceted Knowledge Graph Embeddings for Natural Language Processing. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*. 2017 (Extended abstract)
- W8 Tao Zhou, **Muhao Chen**, Demetri Terzopoulos, Jie Yu. Attention-based Natural Language Person Retrieval. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. IEEE, 2017.
- W9 Shi Gao, **Muhao Chen**, Maurizio Atzori, Carlo Zaniolo. SPARQL^T and its User-Friendly Interface for Managing and Querying the History of Knowledge Bases. In *the 14th International Semantic Web Conference (ISWC)*, 2015 (demo).

AWARDS

- *Amazon Research Award*. 2026
- *Outstanding Paper Award*. EMNLP 2024
- *Outstanding Paper Award*. EMNLP 2023
- *Amazon Research Award*. 2023
- *Keston Exploratory Research Award*. 2023
- *Amazon Research Award*. 2022
- *Cisco Faculty Research Award*. 2022
- *NSF CISE Research Initiation Initiative (CRII) Award*. 2021
- *ACM SIGBio Best Student Paper Award*. 2020
- *UCLA Dissertation Fellowship*. 2018-2019
- *Tung OOCL Scholarships*. Tung's Foundation of Hong Kong & The Oriental Overseas Container Line. 2012, 2013
- *Wang-Dao Fellowship*. President of Fudan University. 2013
- *Chun-Tsung Fellowship*. Hui-Chun Chin and Tsung-Dao Lee Chinese Undergraduate Research Endowment. 2012

FUNDING

- NSF / NIH R01: SCH: Proof-Certified Cancer Registry Extraction** 2026-2030
- \$296K my share. (Co-PI)
- Amazon Research Award: Evolutionary Red-Teaming for Safe Multimodal Web Agents** 2026-2027
- \$35K unrestricted fund + \$30K AWS credits. Fall 2025 call.
- Noyce Initiative: Efficient and Scalable Cyber Defenses with Specialized Agents, Models, and Benchmarking** 2026-2027
- \$130K unrestricted fund.
- NSF CICI: UCSS: Enabling a Safe and Directive Multi-modal Foundation Model Ecosystem for Food Science Research** 2025-2028
- \$200K my share. (Co-PI)
- NVIDIA Academic Grant: Robustness Development of World-Model Agents**
- Hardware equipment gift of four RTX PRO 6000 Blackwell Server Edition
- Uniphore Orby Research Grant: On Safety Research for Agentic and Omni-modal AI** 2025-2026
- \$80K unrestricted fund.
- Amazon Trusted AI Prize** 2024-2026
- \$250K research fund. (PI)
- DARPA FoundSci: MetaScientist: Bioinspired Meta-Material Design with an Autonomous Scientist** 2024-2025
- \$180K my share. (Co-PI)
- Amazon Research Award: Robust (Controlled) NLG with Structure-Aware Equivariance Learning** 2023-2024
- \$70K unrestricted fund + \$40K AWS credits. Spring 2023 call.

NSF Proto-OKN: *Knowledge Graph Construction for Resilient, Trustworthy, and Secure Software Supply Chains* 2023-2026

- NSF ITE Grant, \$370K my share. (Co-PI)

Amazon Research Award: *On Faithfulness of Information Extraction* 2023-2024

- \$73.8K unrestricted fund + \$20K AWS credits. Spring 2022 call.

Keston Exploratory Research Award: *Multi-document Newsworthy Event Monitoring and Forecasting* 2023-2024

- \$100K unrestricted fund from Keston Family Foundation. (PI)

Cisco Research Award: *Robust Knowledge Extraction from Text* 2022-2023

- \$70.4K unrestricted fund.

DARPA KMASS Phase 1: *Knowledge Needed in Context* 2022-2023

- \$308K my share. (Co-PI)

Improved Performance, Analytics and Summarization of Synergistic Anticipation of Geopolitical Events 2022-2023

- ARLIS research grant. \$50K my share. (Co-PI)

NSF CRII: III: *Knowledge Graph Completion with Transferable Representation Learning* 2021-2023

- NSF IIS Grant, \$175K. (PI)

DARPA MCS: *Multi-modal Open World Grounded Learning and Inference* 2020-2023

- \$600K my share. (Co-PI)

TEACHING AND MENTORING

Teaching

ECS 189G: Modern Natural Language Processing (UC Davis)

- Winter 2026, Winter 2027

ECS 271: Machine Learning and Discovery (UC Davis)

- Spring 2024, Spring 2025, Spring 2026, Spring 2027

ECS 289G / ECS 282: Advanced Natural Language Processing (UC Davis)

- Winter 2024, Winter 2025, Fall 2026

CSCI 544: Applied Natural Language Processing (USC)

- Spring 2022

PhD Alumni

Wenxuan Zhou, PhD in Computer Science, USC. 2023

- Dissertation Title: *Robust and Generalizable Knowledge Acquisition from Text*
- Employment: Research Scientist at Google Deepmind

Eric (Ehsan) Qasemi, PhD in Computer Science, USC. 2023

- Dissertation Title: *Multi-modal Preconditioned Inference of Commonsense Knowledge*
- Employment: ML Scientist at LinkedIn

Peifeng Wang, PhD in Computer Science, USC. 2023 (Co-advised with Xiang Ren)

- Dissertation: Externalized Reasoning in Language Models for Scalable, Trustworthy AI
- Employment: Research Scientist at Salesforce AI

Fei Wang, PhD in Computer Science, USC. 2025

- Dissertation: Towards robust and controllable (multimodal) large language models
- Viterbi Honors Program; **USC CS Departmental Best Research Award; Annenberg Fellow. Amazon ML Fellowship.**
- Employment: Research Scientist at Google

Nan (Nancy) Xu, PhD in Computer Science, USC. 2025 (Co-advised with Xuezhe Ma)

- Dissertation: Building trustworthy LLMs: ensuring inference-time coherence and factuality.
- Employment: Research Scientist at Google Deepmind

James Y. Huang, PhD in Computer Science, USC. 2026

- Employment: Senior Applied Scientist at Microsoft

Bangzheng Li, PhD in Computer Science, UC Davis. 2026

- **USC Provost PhD Fellow.** (Transferred to UC Davis in 2023)
- Employment: Research Scientist at Luma.ai

Chen, Muhao – CV (9/10/2026)

Kyle Chickering, PhD in Applied Mathematics, UC Davis. 2026

- Employment: Research Scientist at Luma.ai

Tenghao Huang, PhD in Computer Science, USC. 2026

- Employment: Assistant Professor of Computer Science at University of Utah.

Hadi Askari, PhD in Computer Science, UC Davis. 2026

- Employment: Senior MLE at NVIDIA.

Lab Members/Students

Ph.D. Students:

Boyu Zhu, PhD in Computer Science.	Fall 2026-date
Tinghui Zhu, Ph.D. Student in Computer Science.	Fall 2025-date
Yuankai Li, Ph.D. Student in Computer Science.	Fall 2025-date
Dongwon Jung, Ph.D. Student in Computer Science.	Fall 2024-date
Xiaofei Wen, Ph.D. Student in Computer Science	Fall 2024-date
Jacky Mo, Ph.D. Student in Computer Science (Co-advised with Zhe Zhao)	Fall 2024-date
Rui Cai, Ph.D. Student in Computer Science (Co-advised with Zhe Zhao)	Fall 2024-date
Qin (Jacqueline) Liu, Ph.D. Student in Computer Science.	Fall 2022-date

M.S. and Undergraduate Students:

Keming (Luke) Lu, MS Student in Industrial and System Engineering.	Fall 2021-date
Shikhar Singh, MS Student in Computer Science.	Fall 2021-Summer 2022
Tianyi (Lorena) Yan, Undergraduate Student, Computer Science (CURVE Fellowship)	Fall 2022-date
Jacky Mo, Undergraduate Student, Computer Science (CURVE Fellowship)	Fall 2022-Spring 2024
Mingtao Dong, Undergraduate Student, Computer Science (Provost's Research Fellowship)	Spring 2022-date
Jiashu Xu, Undergraduate Student, Computer Science (CURVE Fellowship)	Fall 2021- Spring 2022
Zhewei Xu, Undergraduate Student, Computer Science (CURVE Fellowship)	Fall 2021-Spring 2022
Logan Norman, Undergraduate Student, Computer Science (Viterbi Fellowship)	Fall 2021-Spring 2022

PROFESSIONAL SERVICE

Society Organization

NAACL Executive Board Member, 2026-2027.

ACL Special Interest Group on NLP Security (SIGSEC) -- Founding Officer and Secretary, 2023.9-date

Conference Organization/Senior Committee Member:

2027: NAACL (Senior Area Chair), ICLR (Area Chair)
2026: ACL (Student Volunteer Chair; Area Chair), ICLR (Area Chair), EMNLP (Area Chair), NeurIPS (Area Chair)
2025: NAACL (Area Chair), ACL (Area Chair), EMNLP (Area Chair), ICLR (Area Chair)
2024: NAACL (Area Chair), EMNLP (Area Chair), ACL (Area Chair), LREC-COLING (Area Chair – Information Extraction)
2023: AACL (Area Chair), NAACL (Senior Area Chair – Information Retrieval and Text Mining), ACL (Area Chair), EMNLP (Area Chair – Information Extraction), NLPCC (Area Chair)
2022: AACL (Senior PC), NAACL (Senior Area Chair), NAACL SRW (Faculty Mentor), EMNLP (Area Chair), AACL (Area Chair)
2021: AACL (Senior PC), IJCAI (Senior PC)
2019: IEEE AIKE (Doctoral Consortium Chair)

Workshop Organization Member:

AAAI spring Symposium on AI for Engineering and Scientific Discoveries, 2025
LLMSEC@ACL, 2025
AI4Research@IJCAI, 2024
Indirect, Weak and Self Supervision for Knowledge Extraction (Wise-Supervision@AKBC), 2022
Deep Learning on Graphs for Natural Language Processing (DLG4NLP@NAACL), 2022

PC Member:

Chen, Muhao – CV (9/10/2026)

2021: ACL, EACL, NAACL, WSDM, WWW.

2020: AAAI, AACL-IJCNLP, AKBC, COLING, EMNLP, IJCAI, ISWC, KDD, SIGIR, WSDM, *SEM.

2019: AAAI, AKBC, NAACL, ACL, EMNLP-IJCNLP, BigData, NLPCC, WISE, ICSC.

2018: AAAI, EMNLP, BigData, NLPCC, SoCal NLP.

Editorial Board: Frontiers in Big Data

Journal Reviewer: TPAMI, TACL, AIJ, TNNLS, TASLP, Pattern Recognition, Bioinformatics, PLOS Computational Biology, Briefings in Bioinformatics, Cell Systems, Clinical and Translational Medicine, Comput. & Struct. Biotechnol., TKDD, TOIS, W3J, TKDE, TII, BMC Genomics, BMC Medical Genomics, BMC Human Genetics, Human Genomics, Quantitative Biology, GeoInformatica, Information Sciences, Nature Machine Intelligence.

Panelist:

2026: DoE Panel

2025: DoE Panel

2024: DoE Panel

2021, 2022: NSF CISE Core Panels

University Service

Department of Computer Science, UC Davis CS Colloquia Organizer	2025.1-date
Departments of Linguistics and Statistics, UC Davis Faculty Hiring Committee	2025
UC Davis, Fellowship Review Committee	2024
UC LFRP, Planning Committee	2024
Department of Computer Science, UC Davis Faculty Hiring Committee	2024
NSF REU Site, Co-organizer	2022, 2023
USC ISI Institutional AI Seminar, Organizer	2021.7-2022.12
Department of Computer Science, USC Faculty Hiring Committee	2021, 2022
Viterbi School of Engineering, USC Fellowship Committee	2021
Department of Computer Science, USC PhD Admission Fellowship Committee	2021

PRESENTATIONS

Invited talks, colloquia and tutorials

1. Reasoning Guardrails for the Agentic Web. *Invited talk, Amazon AGI. July 2026*
2. Reasoning Guardrails for the Agentic Web. *Invited talk, Amazon Search. June 2026*
3. Reasoning Guardrails for the Agentic Web. *Invited talk, USF Rising Innovators Talk. Jan 2026*
4. Reasoning Guardrails for the Agentic Web. *Invited talk, Agentic AI Frontier Seminars. Nov 2025*
5. Reasoning Guardrails for the Agentic Web. *Invited talk at Workshop on Agentic AI, CoLM. Oct 2025*
6. Reasoning Guardrails for the Agentic Web. *Invited talk, Davis Linguistics Colloquium. Oct 2025*
7. Mitigating Backdoor Threats to Large Language Models. *EECS Colloquium, UC Merced. Nov 2024*
8. Mitigating Backdoor Threats to Large Language Models. *Invited talk, Allerton Conference. Sept 2024*
9. Mitigating Backdoor Threats to Large Language Models. *Invited talk, Adobe. Aug 2024*
10. Combating Security and Privacy Issues in the Era of Large Language Models. *Half-day Tutorial at NAACL. June 2024*
11. Indirectly Supervised Natural Language Processing. *Half-day Tutorial at ACL. July, 2023.*
12. Robust and Indirectly Supervised Knowledge Acquisition. *CS Research Colloquium, ASU. March 2023*
13. Robust and Indirectly Supervised Knowledge Acquisition. *CS Research Colloquium, UMN. March 2023*
14. Robust and Indirectly Supervised Knowledge Acquisition. *CS Research Colloquium, UC Davis. Feb 2023*
15. Robust and Indirectly Supervised Knowledge Acquisition. *CS Research Colloquium, UCSB. Feb 2023*
16. Robust and Indirectly Supervised Information Extraction. *Invited talk, AI & ML Seminar, UCI. Nov 2022*

17. Robust and Indirectly Supervised Information Extraction. *Invited talk, NL Seminar, UMN*. Nov 2022
18. Robust and Indirectly Supervised Information Extraction. *Invited talk, CS Department Seminar (CS201), UCLA*. Oct 2022.
19. Robust and Indirectly Supervised Information Extraction. *Invited talk, CS Department Seminar (remote), Nanjing University*. Sept 2022.
20. Robust and Indirectly Supervised Information Extraction. *Invited talk, Apple, San Jose*. Aug 2022.
21. Robust and Indirectly Supervised Information Extraction. *Invited talk, Microsoft CSR Distinguished Talk Series*. July 2022.
22. New Frontiers of Information Extraction. *Half-day Tutorial at NAACL*. July, 2022.
23. Understanding Event Processes in Natural Language. *Invited talk, NLP Seminar, UC Santa Cruz*. April 2022.
24. Understanding Event Processes in Natural Language. *Invited talk, CS Department Seminar, Rutgers–New Brunswick*. March 2022.
25. Understanding Event Processes in Natural Language. *Invited talk, ML Seminar, Purdue University*. Sept 2021.
26. Understanding Event Processes in Natural Language. *Invited talk, Tencent AI Lab, Seattle*. Sept 2021.
27. Understanding Event Processes in Natural Language. *Invited talk, University of Central Florida*. Aug 2021.
28. Understanding Event Processes in Natural Language. *Invited talk, IBM Research Almaden*. July 2021.
29. Understanding Event Processes in Natural Language. *Invited talk, CS Department Seminar, Fudan University*. June 2021.
30. Understanding Event Processes in Natural Language. *Invited talk, CS Department Seminar, Nanjing University*. June 2021.
31. Understanding Event Processes in Natural Language. *Invited talk, NLP Seminar, National University of Singapore*. May 2021.
32. Understanding Event Processes in Natural Language. *Invited talk, NLP Seminar, OSU*. May 2021
33. Understanding Event Processes in Natural Language. *Invited talk, CS Research Colloquium, UCSB*. April 2021
34. Understanding Event Processes in Natural Language. *Invited talk, Frontier Topics in Vision and Language, ASU*. Mar 2021.
35. Understanding Event Processes in Natural Language. *Invited talk at Machine Learning and Big Data Seminar, UCLA*. Nov 2020
36. Event-Centric Natural Language Processing. *Half-day Tutorial at ACL*. Aug, 2021.
37. From Tables to Knowledge: Recent Advances in Table Understanding. *Half-day Tutorial at KDD*. Aug, 2021.
38. Understanding Event Processes in Natural Language. Understanding Event Processes in Natural Language. *Invited talk at Language Technology Seminar, University of Cambridge*. Nov 2020
39. Event-Centric Natural Language Understanding. *Half-day Tutorial at AAAI*. Feb, 2021.
40. Knowledge Acquisition with Transferable Representation Learning. *CS Research Colloquium, USC*. Nov 2020
41. Knowledge Acquisition with Transferable Representation Learning. *AI Seminar, USC ISI*. Los Angeles, Jan 2020.
42. Knowledge Acquisition with Transferable Representation Learning. *IBM Research Almaden*. San Jose, Jan 2020.
43. Knowledge Acquisition with Transferable Representation Learning. *Invited talk at FDSiF, Fudan Univ., Shanghai, China*. Dec 2019.
44. Knowledge Acquisition with Transferable Representation Learning. *Invited talk at SNAP Seminar, Stanford University*. Jan 2019.
45. Knowledge Acquisition with Transferable Representation Learning. *Machine Learning and Big Data Seminar, UCLA*. Jan 2019.
46. Recent Advances in Transferable Representation Learning. *Half-day tutorial at AAAI*. NYC, NY, Feb 2020.
47. Neural Article Pair Modeling for Wikipedia Sub-article Matching. *Google Search Intelligence Seminar Talk*. Sept 2017
48. Reasoning Across Multiple Spatiotemporal Granularity Systems. *Invited talk at Teradata Labs, El Segundo, CA, USA*. Mar 2015

INDUSTRIAL EXPERIENCE

Microsoft Research, Redmond, WA (NLP Group)

2018.6–2018.9

- **Research Intern**, *Embedding Edge-attributed Relational Hierarchies*. [P135]

Google, Mountain View, CA (Google Knowledge Graph)

2017.6–2017.9

- **Research Intern**, *Neural Article Pair Modeling for Large-scale Sub-article Relation Extraction*. [P143,P140,P127]

Google, Mountain View, CA (Procella Real-time Data Infrastructure)

2016.6–2016.9

Chen, Muhao – CV (9/10/2026)

- **System SDE Intern**, *Dynamic Shard-partitioning for Large-scale Windowed Event Streams*.

Teradata Labs, Los Angeles, CA (Optimizer Group)

2015.6~2015.9

- **R&D Intern**, *Multi-task Learning for Cost-based Database Optimizers*.